



ScopeAngleChecker

User Manual

Contents

1	Introduction	5	Measuring
2	First-Time Setup	6	Using Voice Commands
3	Hospitals and Inspectors	7	Records and Reports
4	Registering Models	8	Settings

01

Introduction

What This App Does

- Just photograph the tip of the endoscope — the bending angle is measured automatically
- Records are saved by hospital, inspector, and scope model
- Results are compared with the specified angle automatically and marked Pass or Needs Attention
- View trends on a graph and export reports as PDF or CSV

IMPORTANT

Please Read First (Important Notes)

- You must keep using the angle measurement plate. Checking with this app does not replace checking with the plate.
- Measured values may deviate depending on shooting conditions (lighting, camera distance and angle).
- This app is not a medical device or diagnostic-support software under Japan's Pharmaceuticals and Medical Devices Act.
- Interpreting the results, and deciding whether to keep using or withdraw an endoscope, is the user's own responsibility.
- Deleting the app, changing phones, or updating iOS can erase your data. Please back up regularly to CSV or PDF.

02

First-Time Setup

What Happens on First Launch

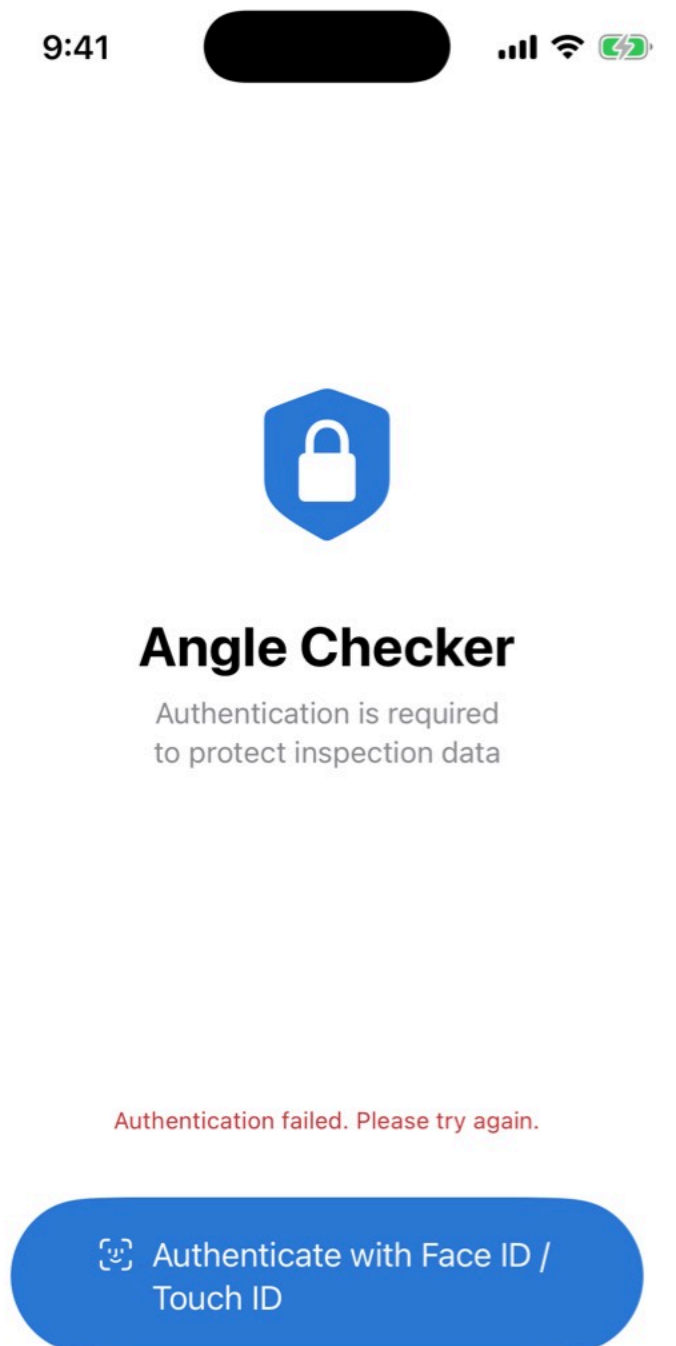
- The first time you open the app, the screens appear in this order.
- The order is fixed; you cannot skip a step.
- 1. Agree to the notes
- 2. How-to tutorial
- 3. Identity check (Face ID / Touch ID)
- 4. Data transfer prompt (choose “Using it for the first time” if this is new)
- 5. Choose a hospital / facility, then an inspector
- → Main screen (five tabs)

NOTE

About the Identity Check

- Face ID / Touch ID starts automatically when the screen opens.
 - If it fails, tap “Authenticate with Face ID / Touch ID” to try again.
 - If the device has no passcode at all, “Data Protection Is Disabled” appears, but you can still continue by choosing “Continue Anyway”.
-
- [Note] You are asked to authenticate every time you switch to another app and come back.

This is intentional, to protect your data — not a malfunction.



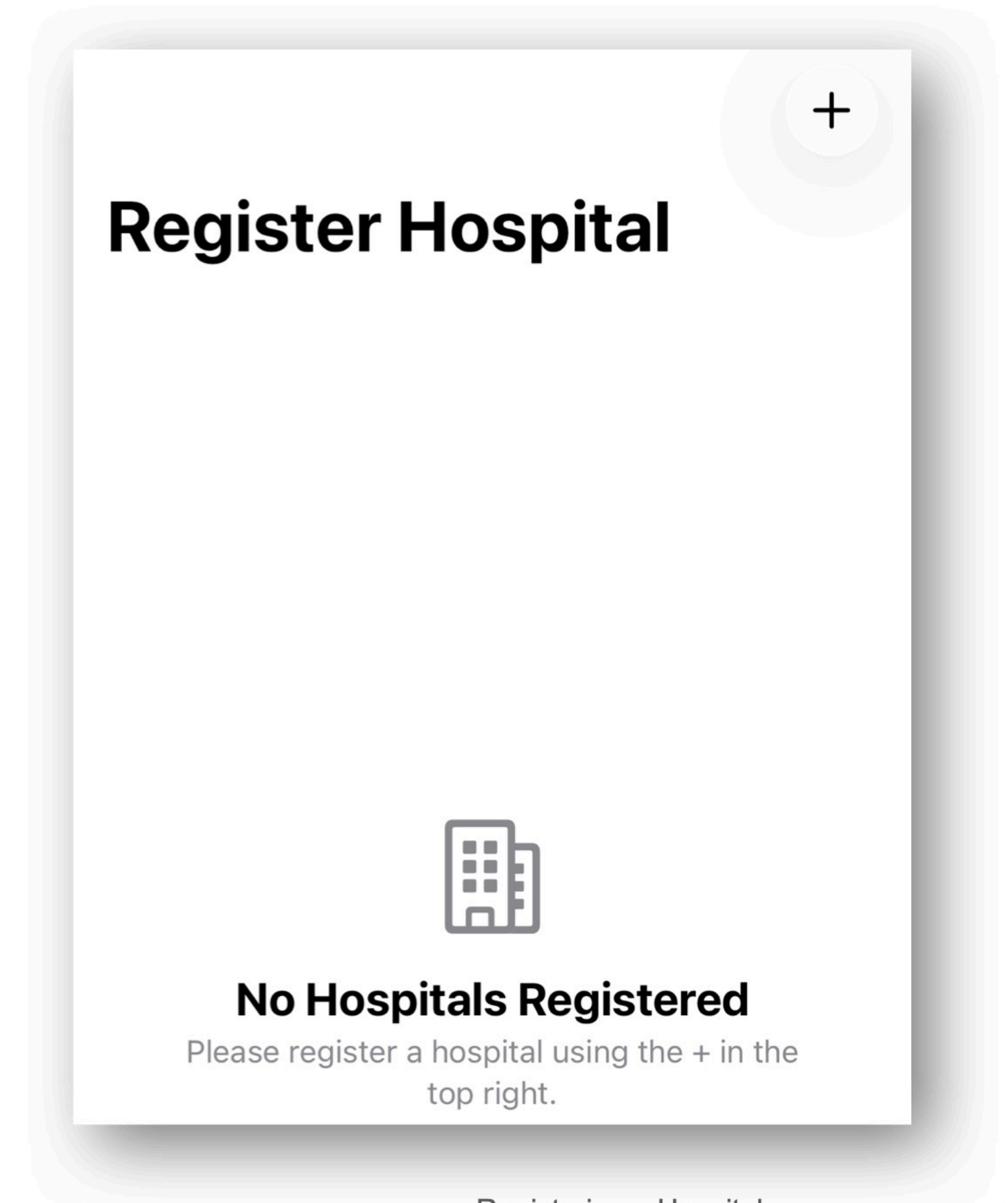
03

Hospitals and Inspectors

STEP 1

Register a Hospital / Facility

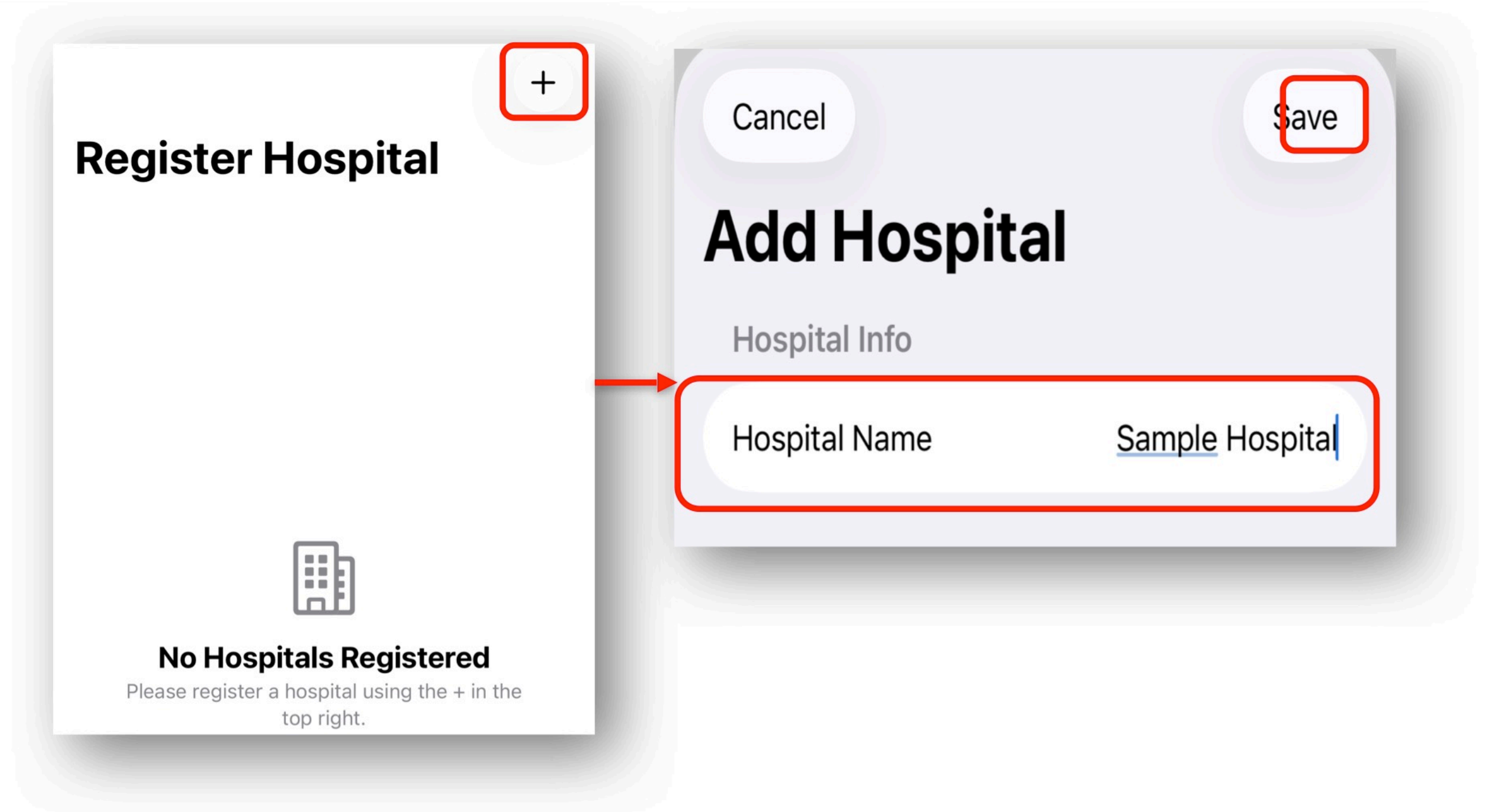
- The first time, no hospital is registered yet, so this screen appears.
- Tap “+” at the top right to add one.
- If a hospital is already registered, just tap it in the list.



STEP 2

Add a Hospital

- The only field is “Hospital Name”.
- Then tap “Save” at the top right.

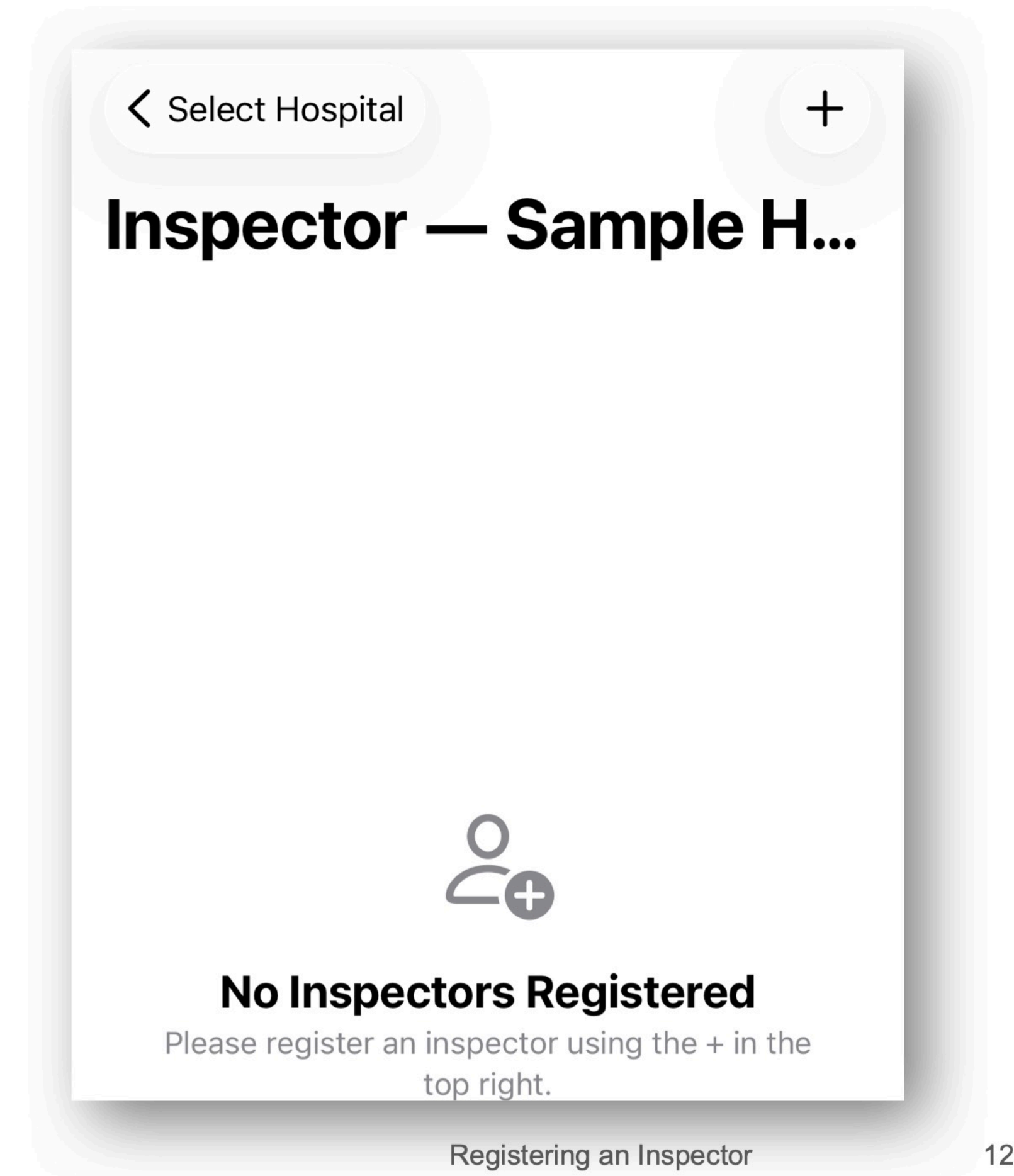


* You can register a clinic or any other facility name — it does not have to be a hospital.

STEP 3

Choose the Inspector (You)

- After you choose a hospital, the inspector screen appears next.
 - The first time, no inspector is registered yet, so this screen appears.
 - Tap “+” at the top right and enter a name.
-
- [Note] The inspector list is shared across the whole app, not kept per hospital. The same list appears whichever hospital you choose.



04

Registering Models

Registering a Model Takes Two Stages

- Registering an endoscope takes two stages.

- 1. Register the model

The specification of that model number (specified angle, tolerance, and so on). Shared by every hospital and facility.

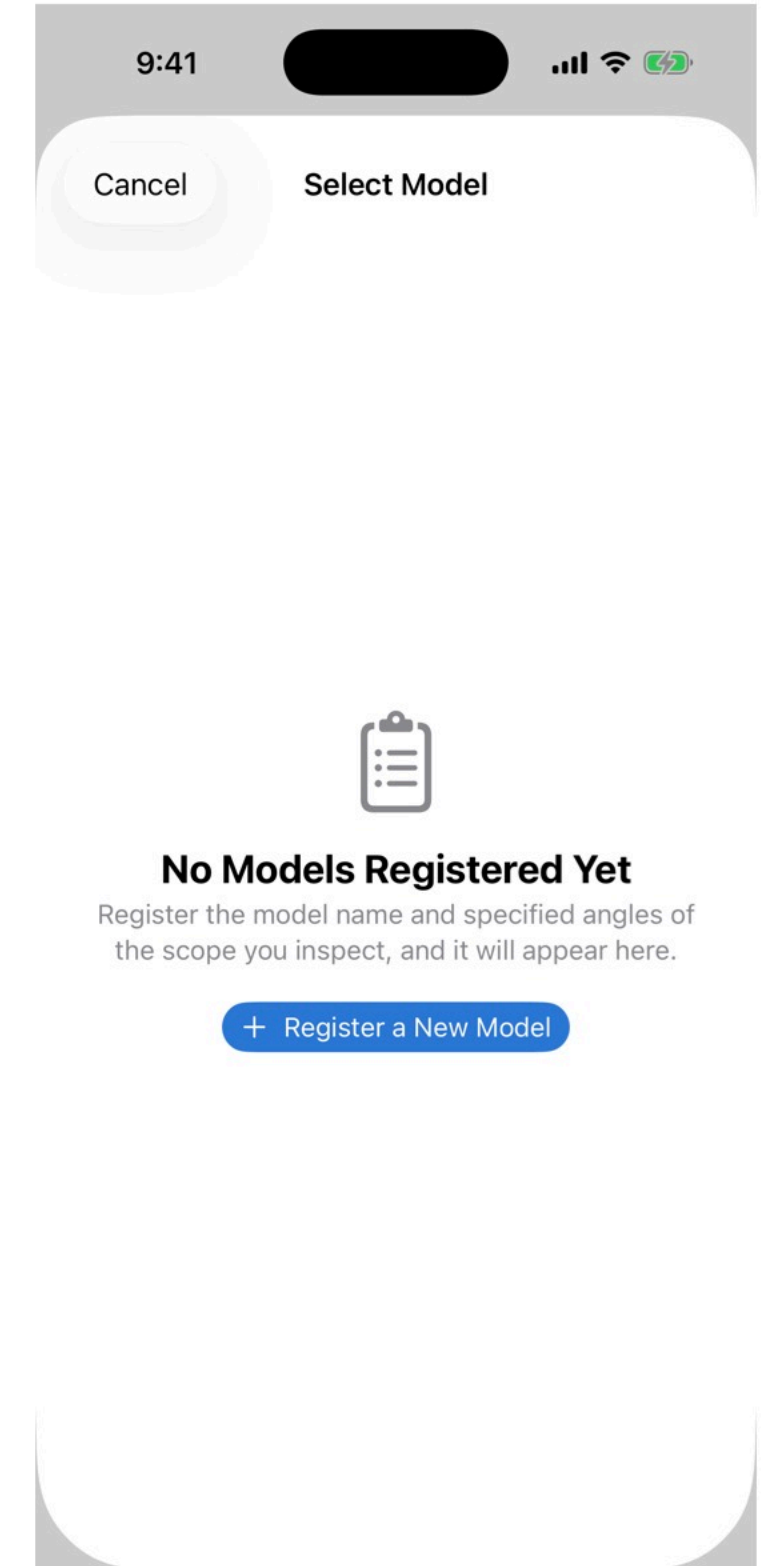
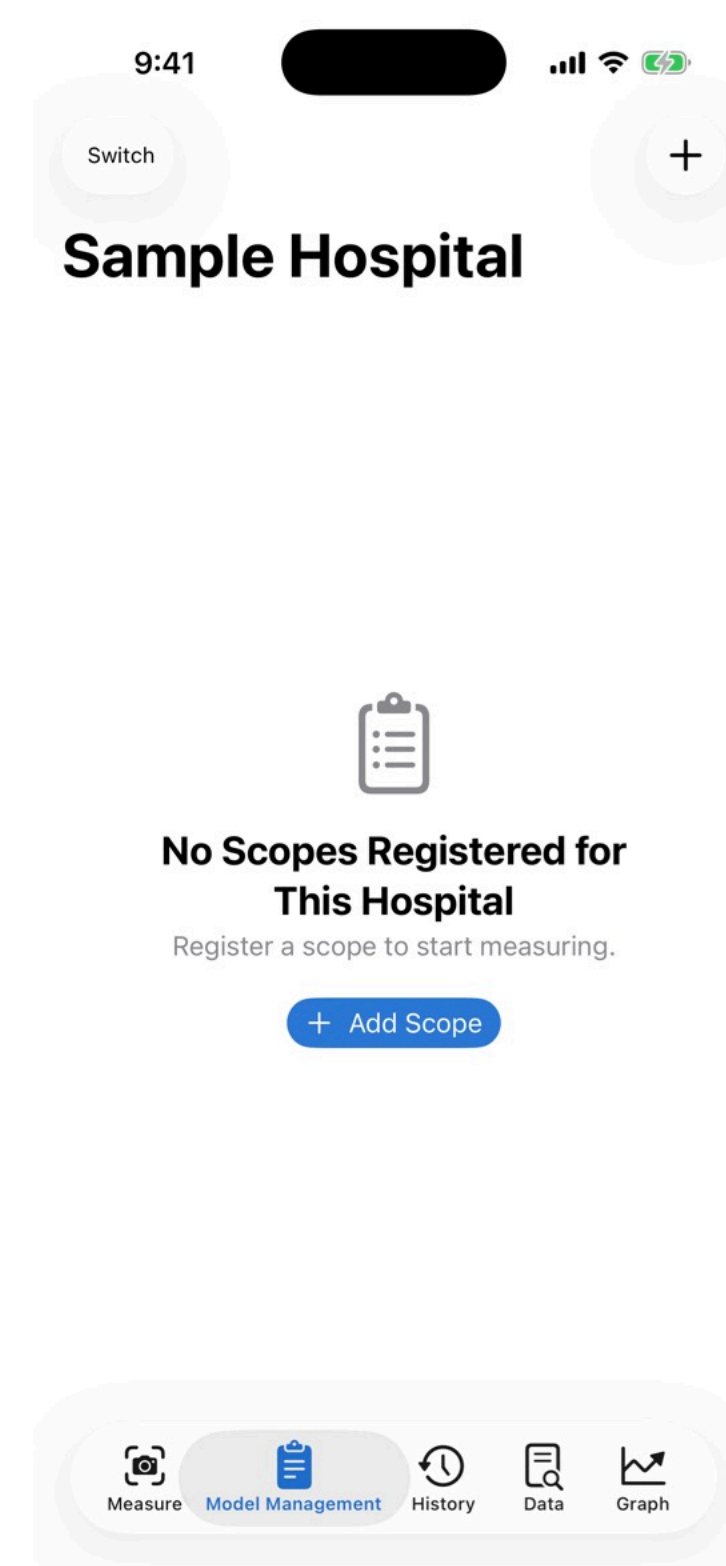
- 2. Add it as a scope at your hospital / facility

The details of the actual unit at that hospital (asset number, serial number, and so on).

STEP 1-1

Choosing or Adding a Model

- Open the “Model Management” tab and tap “+” at the top right.
- Under “Select Model”, choose from the models already registered.
- If there are none yet, tap “+ Register a New Model” to create one.
- You can do the same from Settings (gear) → “Models”.
- This is only needed the first time for a model number. After that you just select it.



STEP 1-2

Entering the Model's Specification

- Tapping “+ Register a New Model” opens this screen.
- Model Name (e.g. XYZ-1000) * required
- Manufacturer (optional) / Category (optional)
- Specified Angle: UP / DOWN / LEFT / RIGHT (optional)
- Tolerance: how many degrees from the specified angle still counts as a pass (default $\pm 30^\circ$)
- Tap “Save” at the top right; you then move straight on to entering the actual unit.

The screenshot shows the 'Register Model' screen in the ScopeAngleChecker app. At the top, there is a status bar with the time 9:41 and battery level. Below the status bar, there is a navigation bar with a back arrow, the title 'Register Model', and a 'Save' button. The screen is divided into several sections: 'Model Info' with fields for 'Model Name' (e.g. XYZ-1000), 'Manufacturer' (Optional), and 'Category' (Not Set); 'Specified Angle (degrees)' with fields for 'UP', 'DOWN', 'LEFT', and 'RIGHT' (all Not Set); and 'Tolerance' with a field set to $\pm 30^\circ$. A note at the bottom states: 'Judged as passing if within this value of the specified angle.'

An Example of a Completed Form

- After filling it in, tap “Save” at the top right to finish.
- This example fills in the specified angle and tolerance as well.
- Once the specified angle is registered, it is used for the pass/fail judgement when measuring and for the pass band on the graph.

<

Register Model

Save

Model Info

Model Name

XYZ-1000

Manufacturer

Sample Optics

Category

Upper GI Endoscope

Specified Angle (degrees)

UP

180

DOWN

180

LEFT

160

RIGHT

160

Tolerance

±

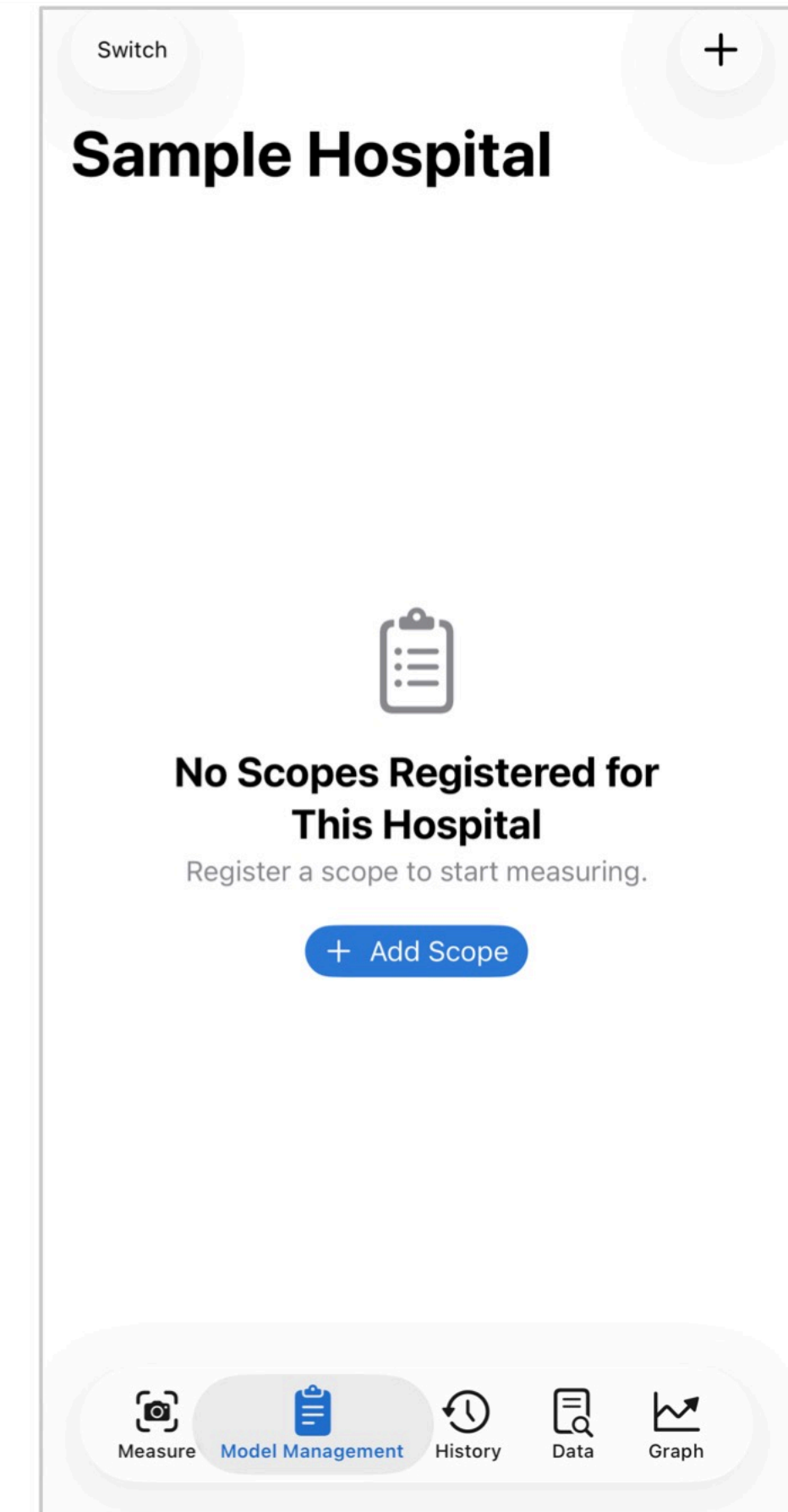
30 °

Judged as passing if within this value of the specified angle.

STEP 2-1

Add It as a Scope at the Hospital

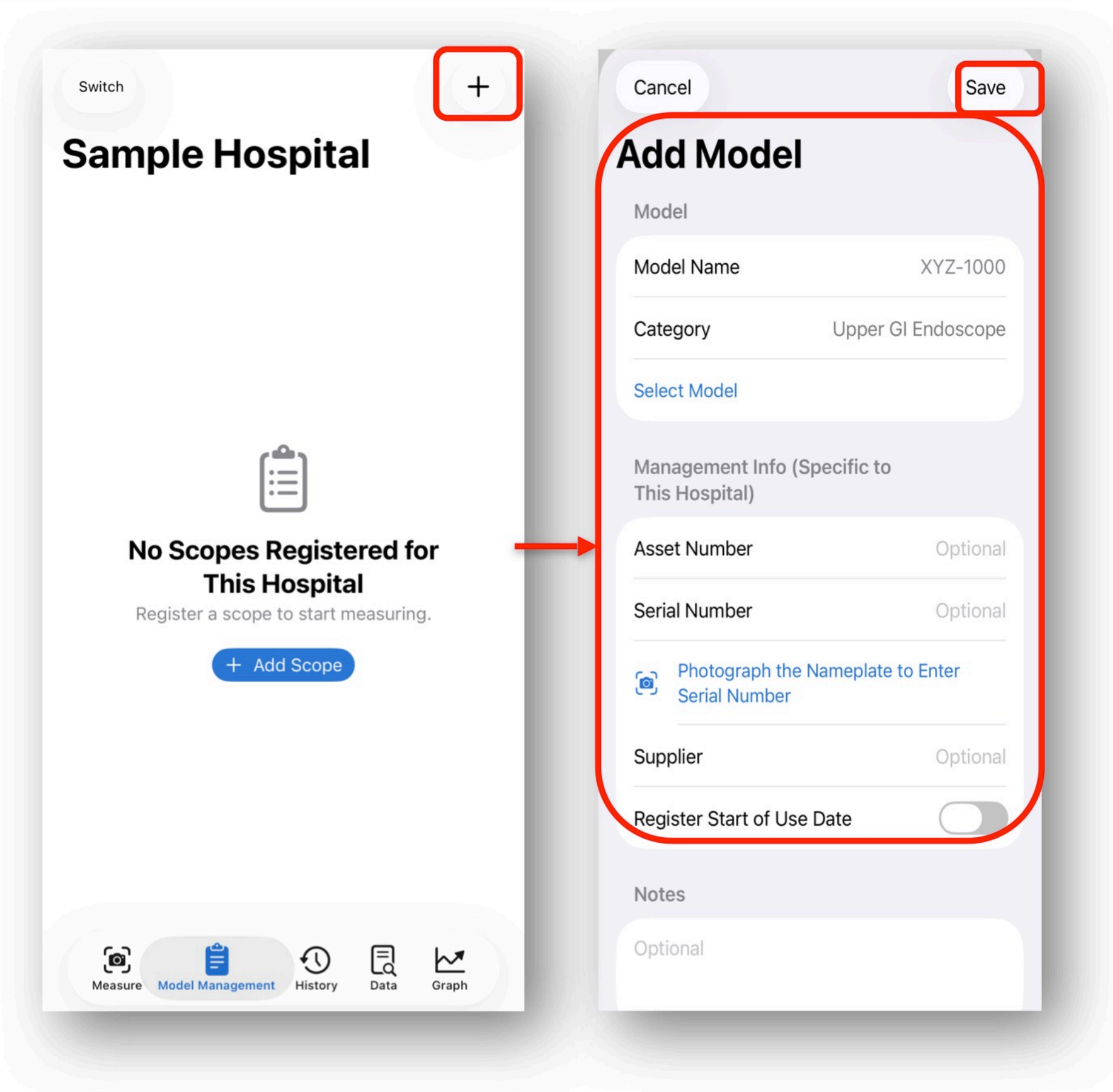
- Open the “Model Management” tab.
- No actual unit is registered at this hospital yet, so this screen appears.
- Tap “+” at the top right to register each unit.



STEP 2-2

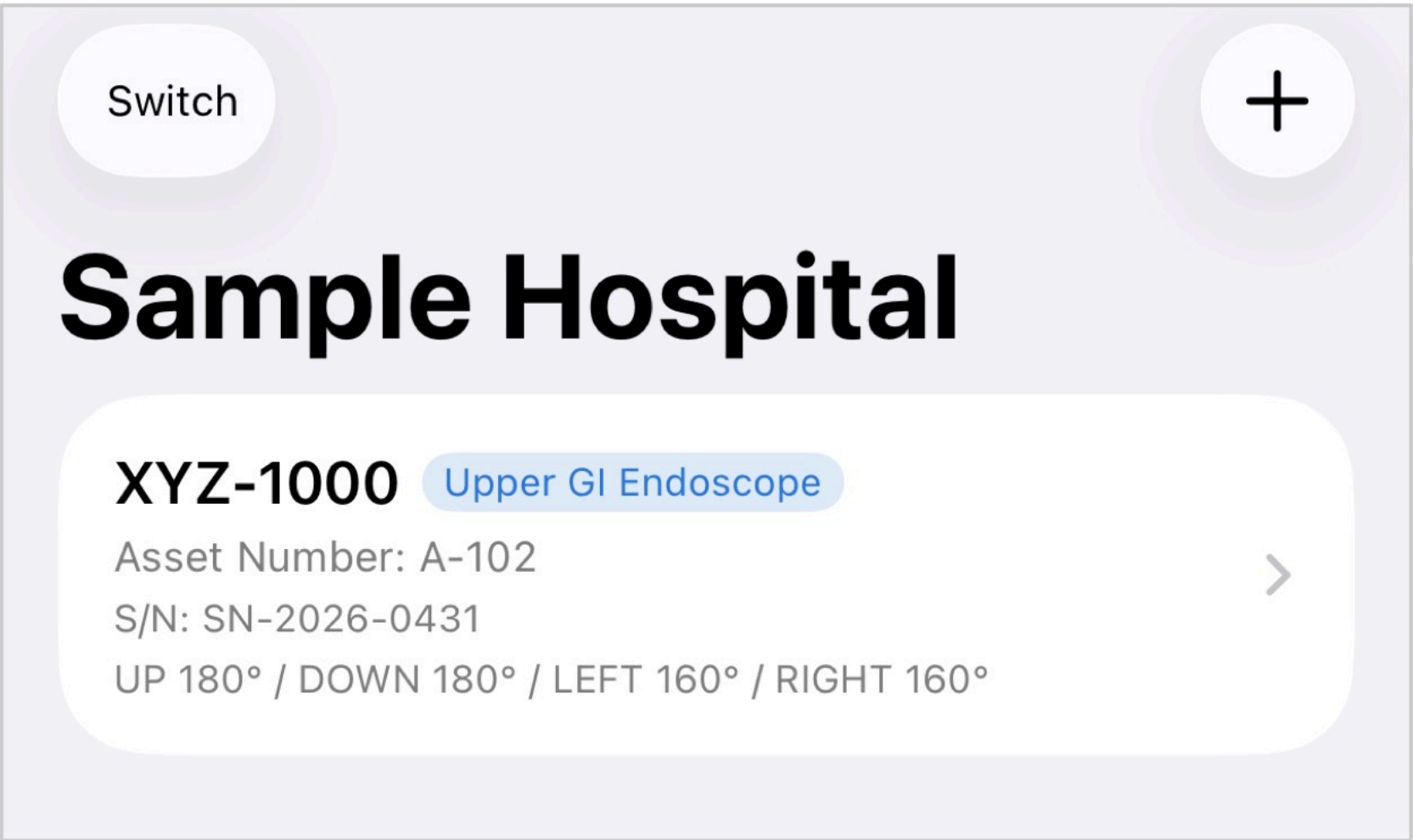
Enter the Unit's Details

- Under “Select Model”, choose the model you registered in stage 1.
- Asset Number (optional)
- Serial Number (optional): type it in, or photograph the nameplate and let the app read it (OCR, processed on the device)
- Supplier / Start of Use / Notes (all optional)



Checking What You Registered

- Once registered, the unit appears in the list on the Model Management tab.
- Model number, category, asset number, serial number, and specified angle are all visible at a glance.
- Tap an entry to edit it; swipe left to delete it.



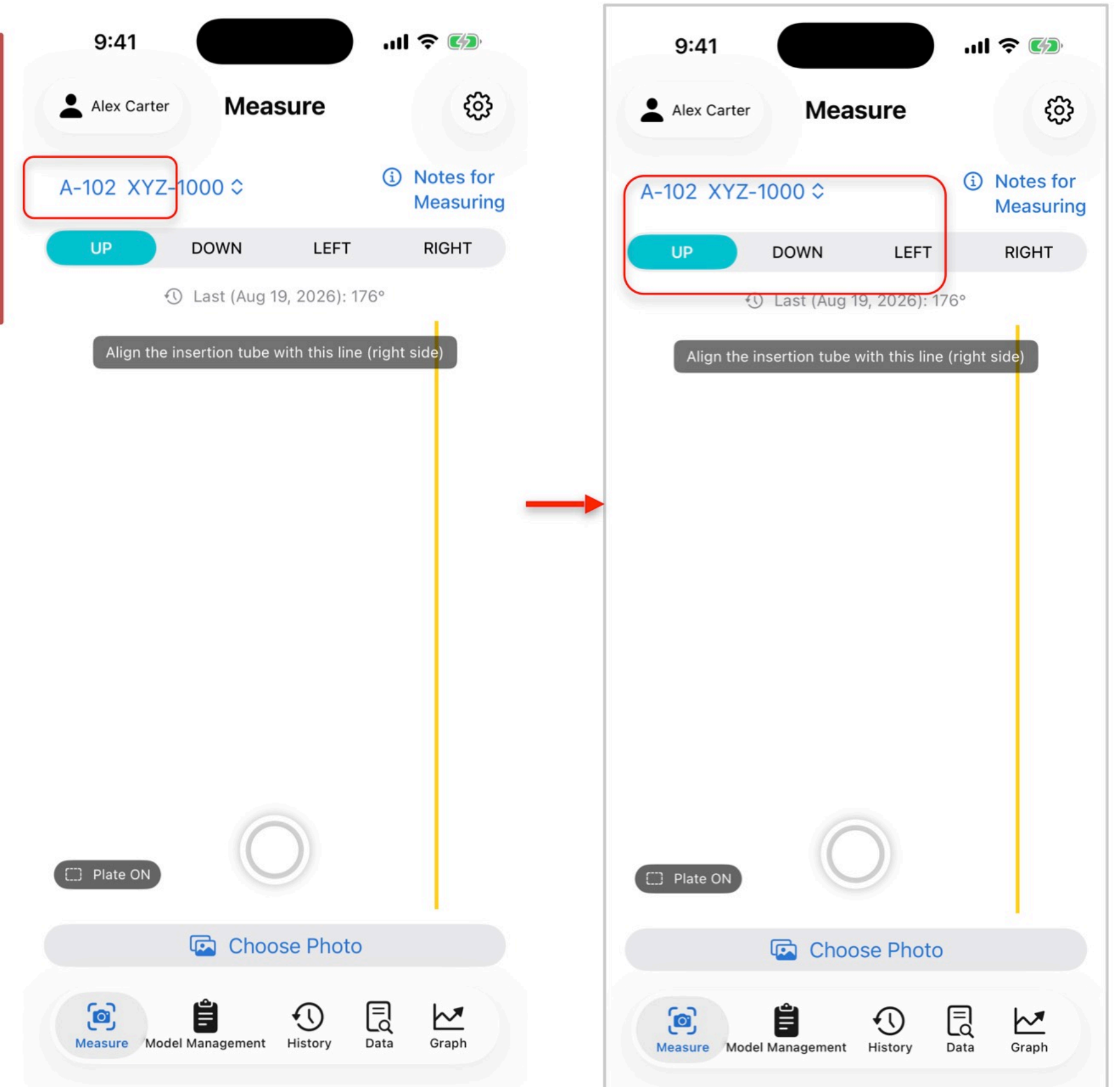
05

Measuring

STEP 1

Choose the Model and Direction

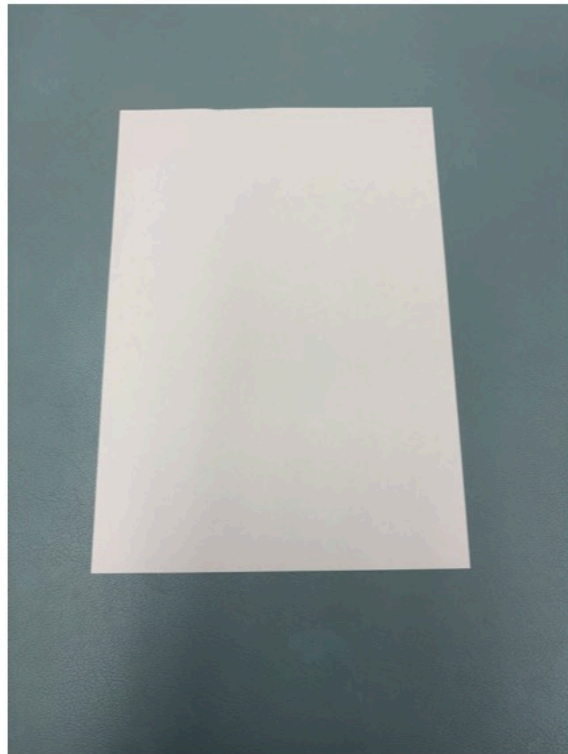
- On the “Measure” tab, choose the model and the direction (UP / DOWN / LEFT / RIGHT).
- If there is a previous record for the same model and direction, “Last 180°” is shown for reference before you shoot.



SET UP

Setting Up Your Shooting Area

- What you need: a sheet of white A4 paper / a rack with holes in its top / a simple light / sticky tape
- The paper goes under the scope; the rack holds the phone, in place of a stand.
- Aim the light so it falls evenly from above and casts no shadow.
- Tape the paper to the table so it cannot shift.



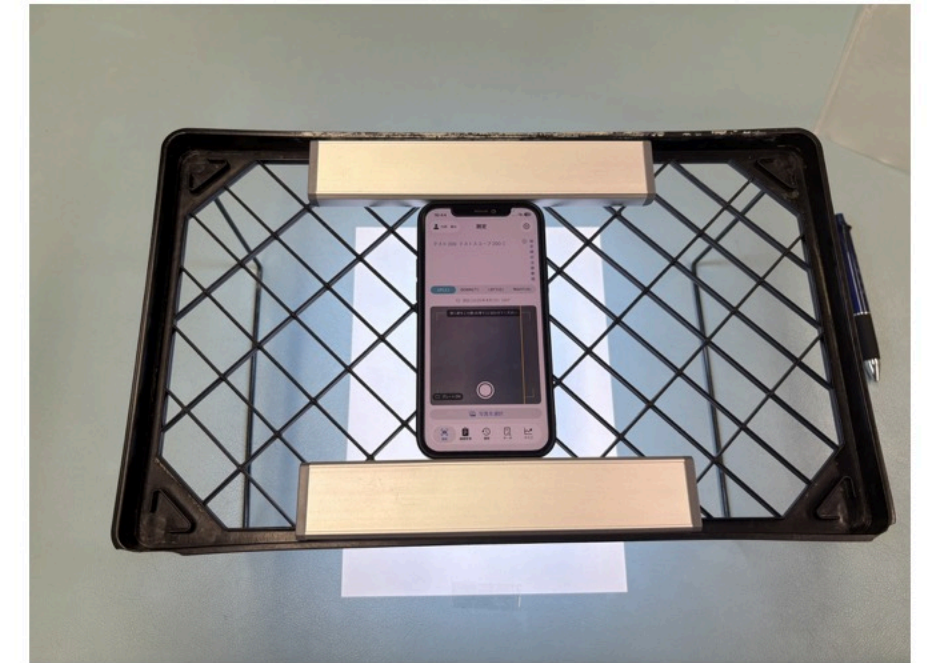
White A4 paper



Rack with a holed top



A simple light



All set up

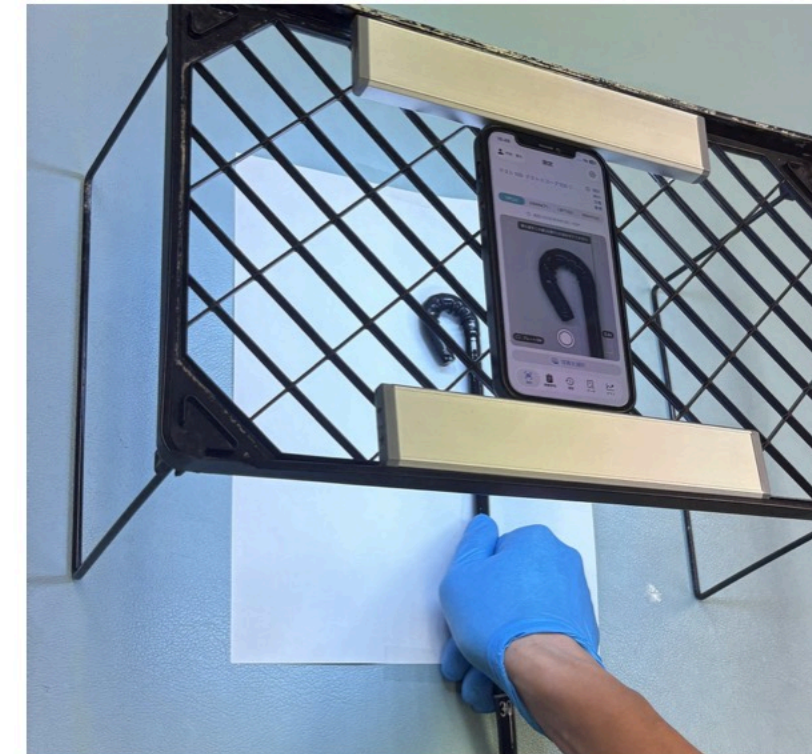
STAND

Shooting on a Stand (recommended)

- Let the camera look through a hole in the rack's top and shoot with the phone resting there — no shake, and the same height and angle every time.
- 1. Tape the A4 paper to the table → 2. Stand the rack over the paper
- 3. Rest the phone so the camera sits over a hole → 4. Switch the light on, lay the scope on the paper, and shoot



Phone resting on the rack



Shooting the scope

Shooting Handheld

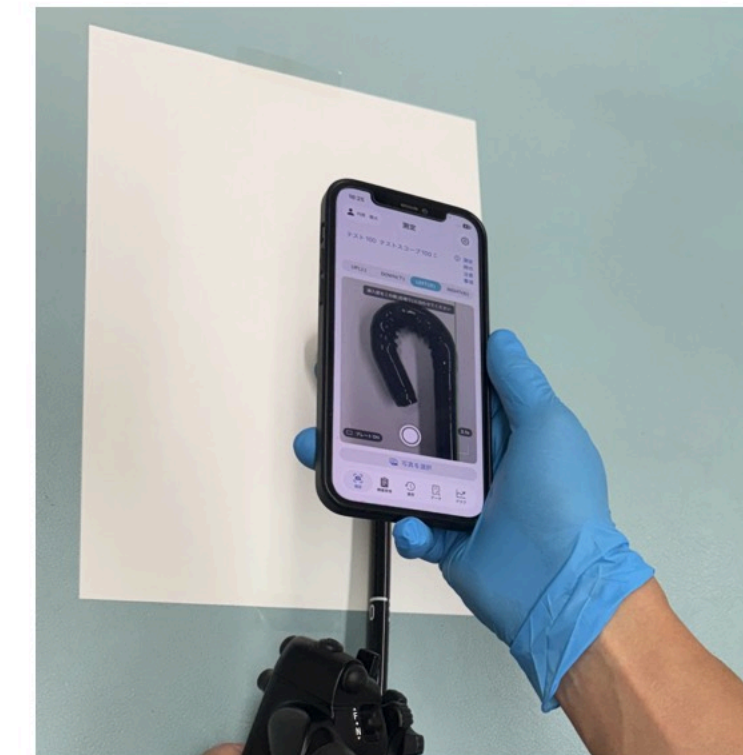
- If you cannot set up a stand, holding the phone by hand still works.
- Settings (gear) → turn on “Handheld mode” to show a level indicator. Centre the bubble and you are shooting from directly above.
- [Important] Rest the scope on the paper. Held up off it, the scope photographs smaller than it is and the angle comes out wrong.



Good — resting on the paper



Bad — the tip is lifted



By hand: hold it directly above

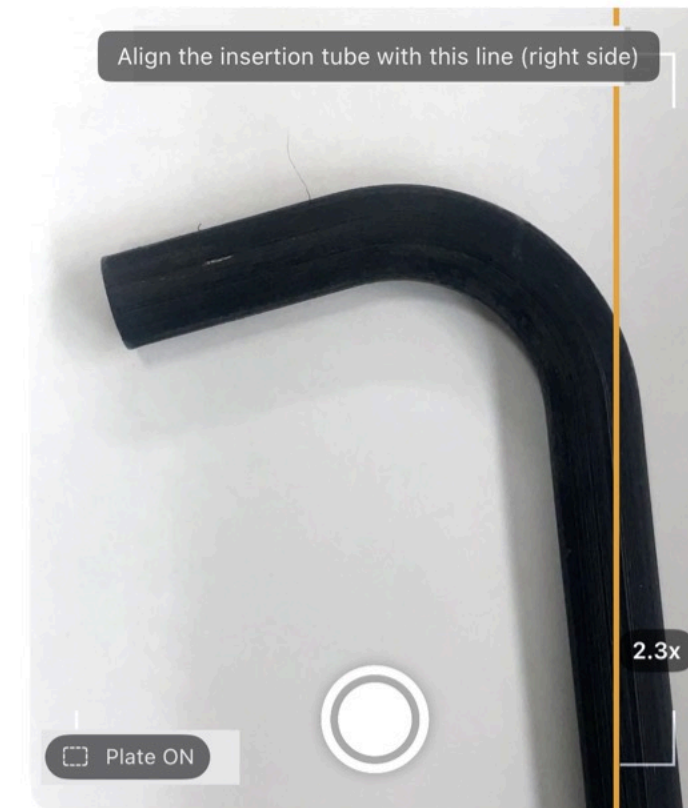
STEP 2

Taking the Photo

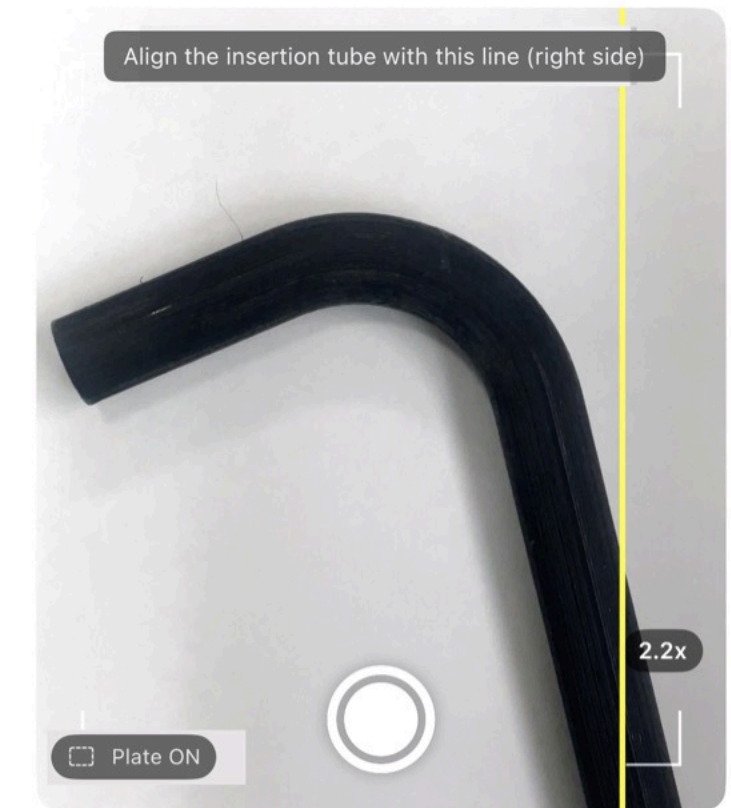
- Lay a sheet of white paper (A4 copy paper works well) underneath and shoot on top of it.
- Rest the scope flat on the paper. Do not hold it up off the paper or let it tilt.
- Line the outer edge of the insertion tube with the reference line on screen (green = good, orange = slightly tilted, yellow = needs adjusting).
- The reference line shifts left or right depending on the direction, to leave room on the side the tip bends toward.
- Zoom until both the tip and the base sit inside the corner guides, then shoot somewhere bright, with no shadows.



Green = good



Orange = slightly tilted



Yellow = needs adjusting

STEP 3

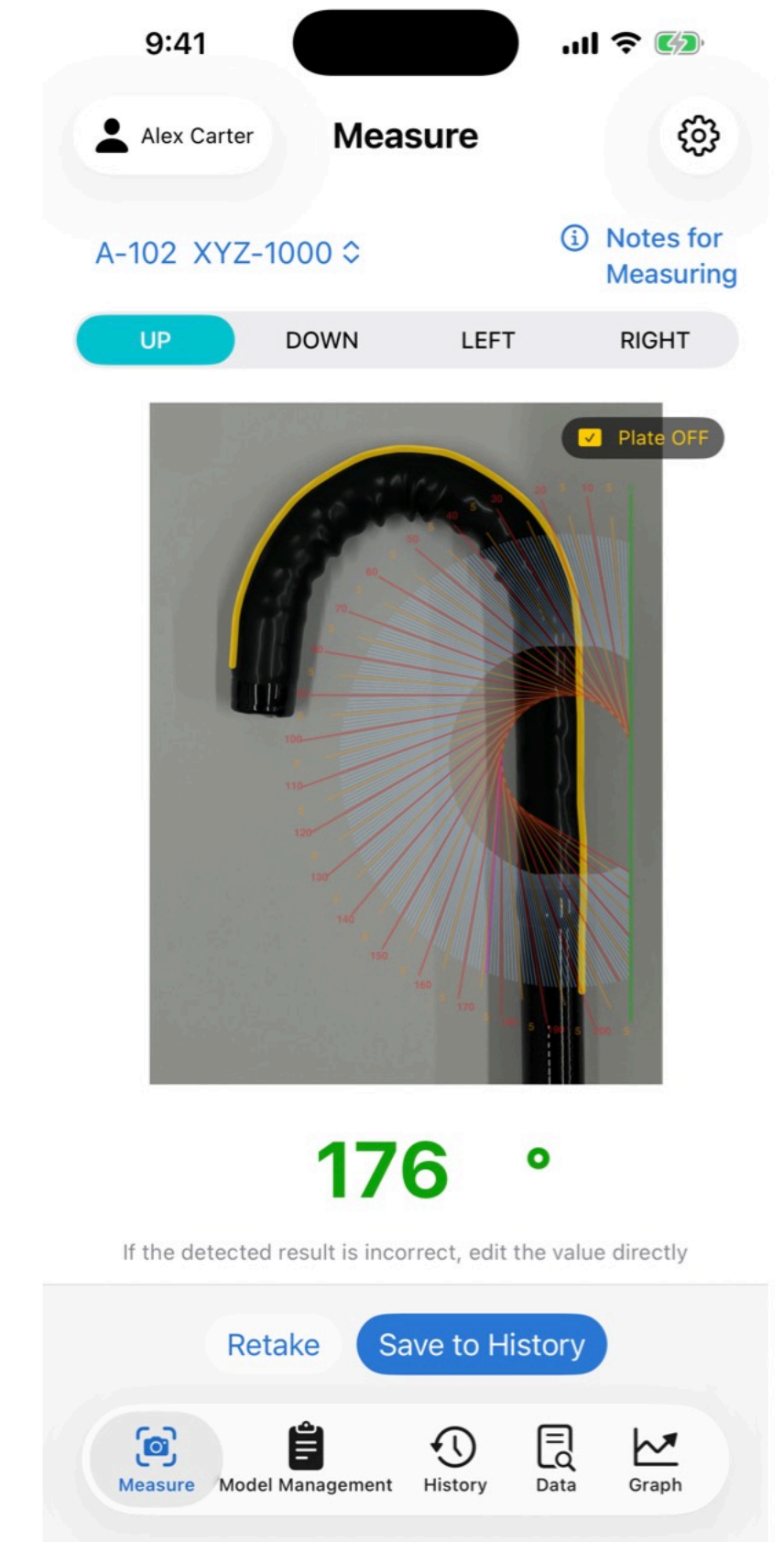
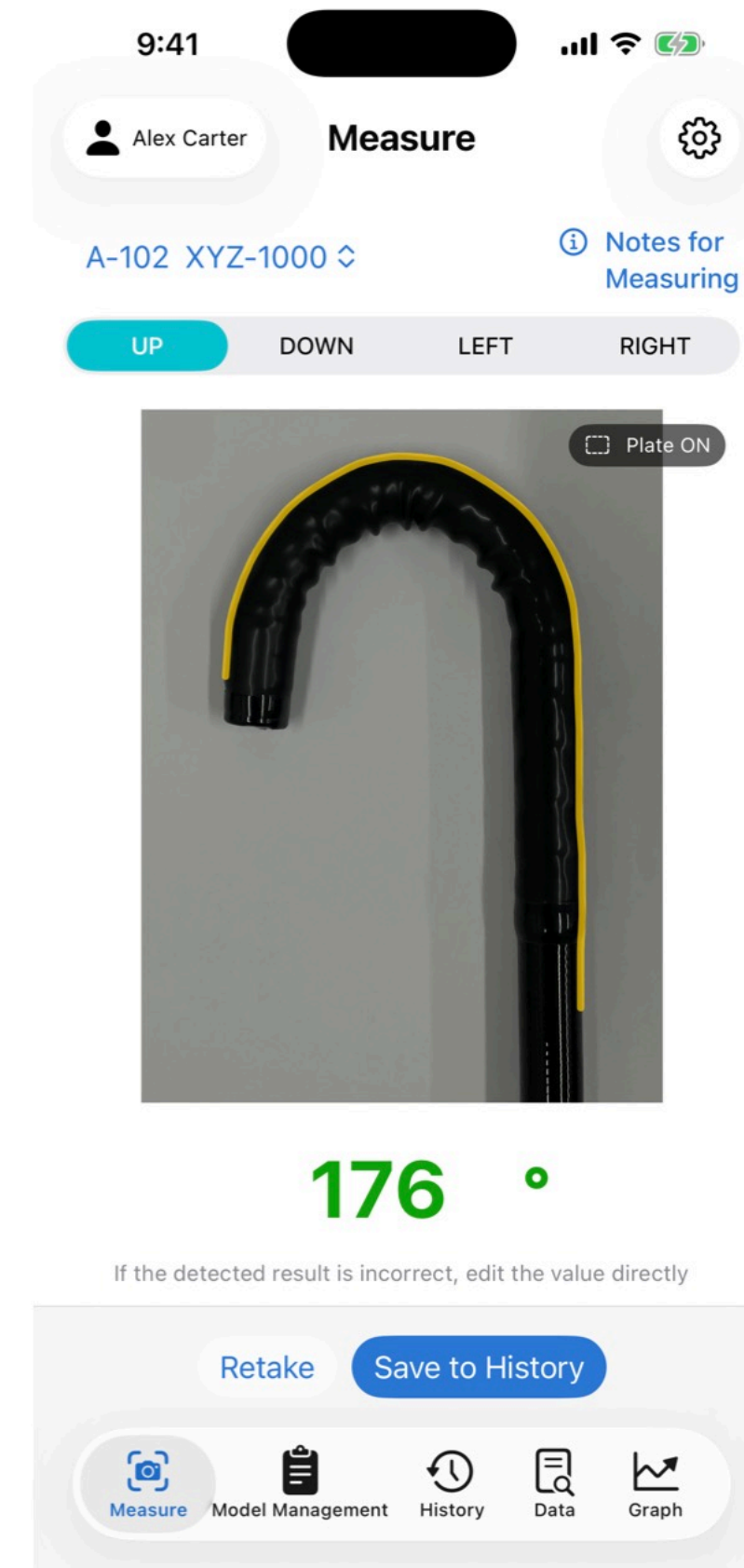
Checking and Saving After the Shot

- After a few seconds the angle is calculated automatically, and the measurement line (yellow) and the value appear.
- If the detection is wrong, tap the value and type the correct one.
- If the shaft is tilted a lot (more than 15°), a warning suggests retaking the photo.
- Check it over and tap “Save to History” (saying “save” out loud works too).
- To shoot again, tap “Retake”.



Overlaying the Angle Plate to Check

- 1. After shooting, tap “Plate ON” at the top right of the photo.
- 2. Drag with one finger to line the centre of the plate up with the base of the scope.
- 3. Pinch with two fingers to match the scale to the real thing (double-tap to reset).
- Only the half of the scale you need for that direction is shown. The line for the measured angle is highlighted in pink.
- * The on-screen plate is for reference only. It does not replace checking with the real angle plate.

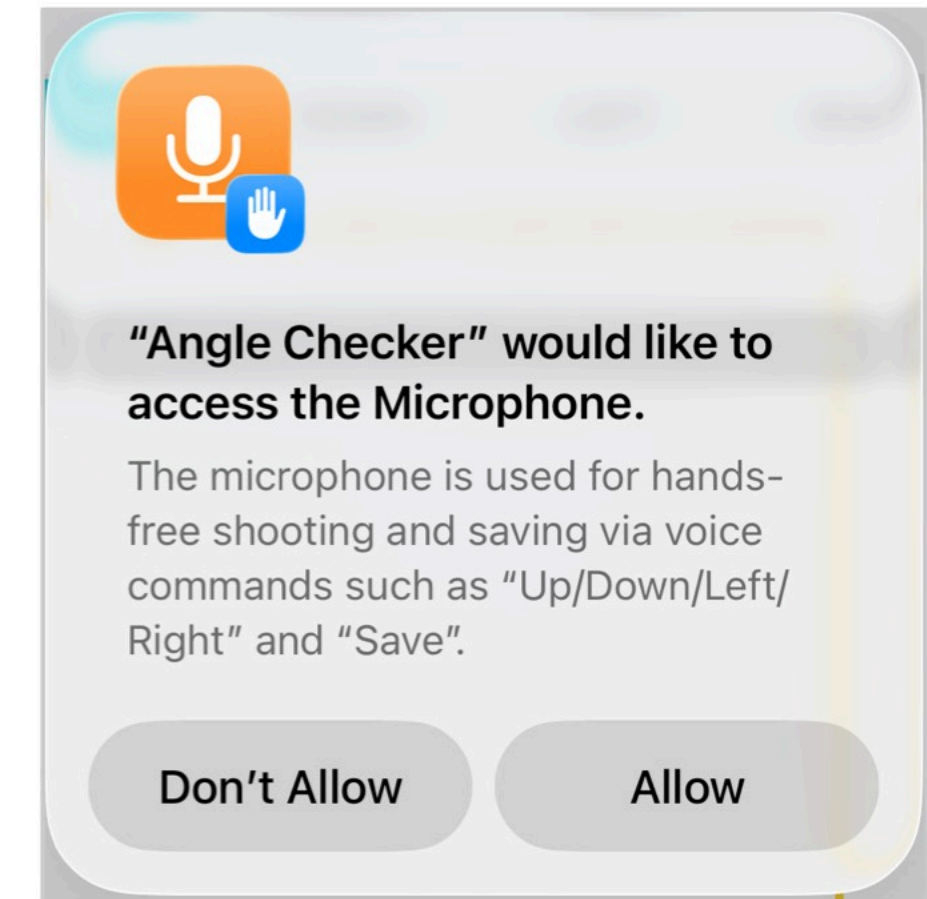
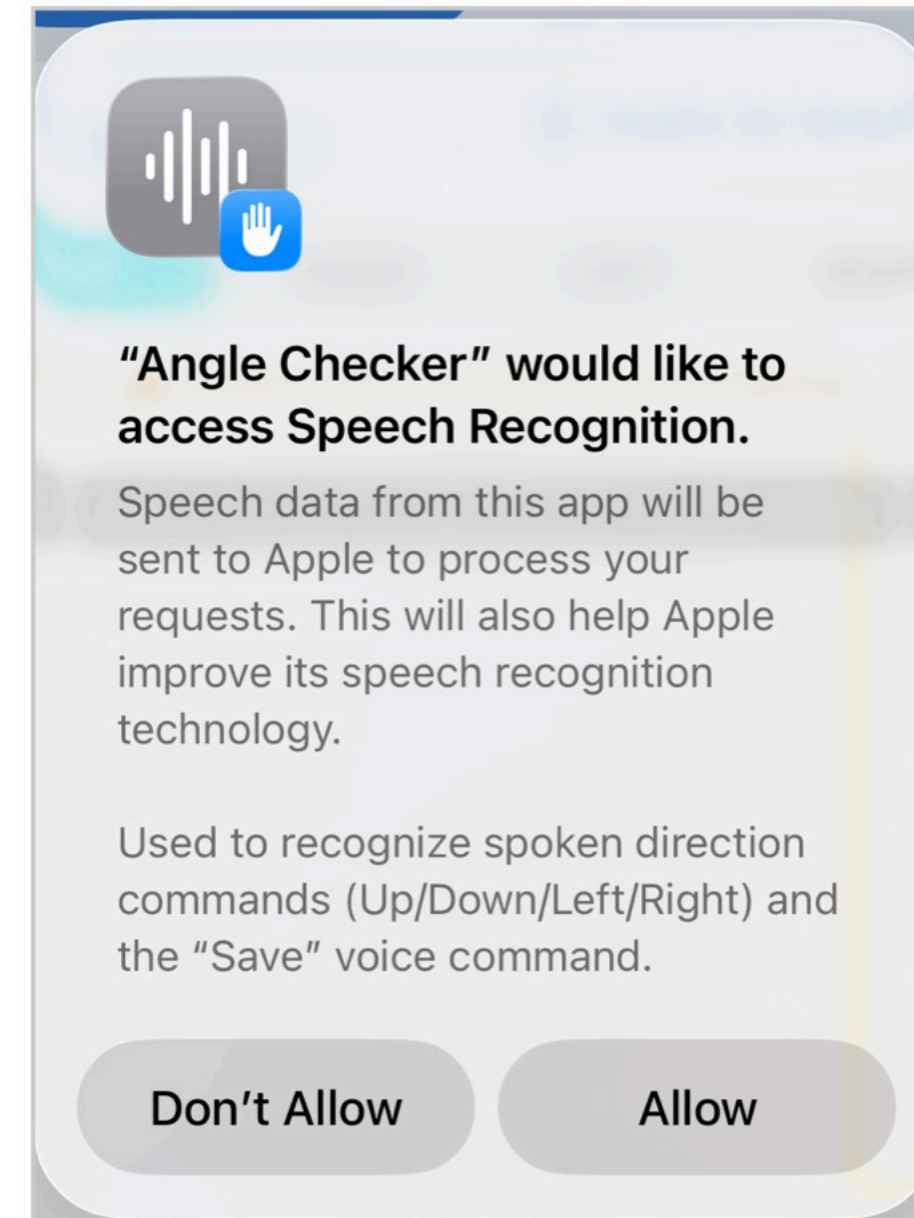


06

Using Voice Commands

Permissions on First Use

- To use voice commands, two permissions are needed the first time.
- 1. Access to speech recognition
- Used to recognise spoken directions (up / down / left / right) and commands such as “save”.
- 2. Access to the microphone
- Used to pick up your voice.
- Please choose “Allow” for both.



The Voice Commands You Can Use

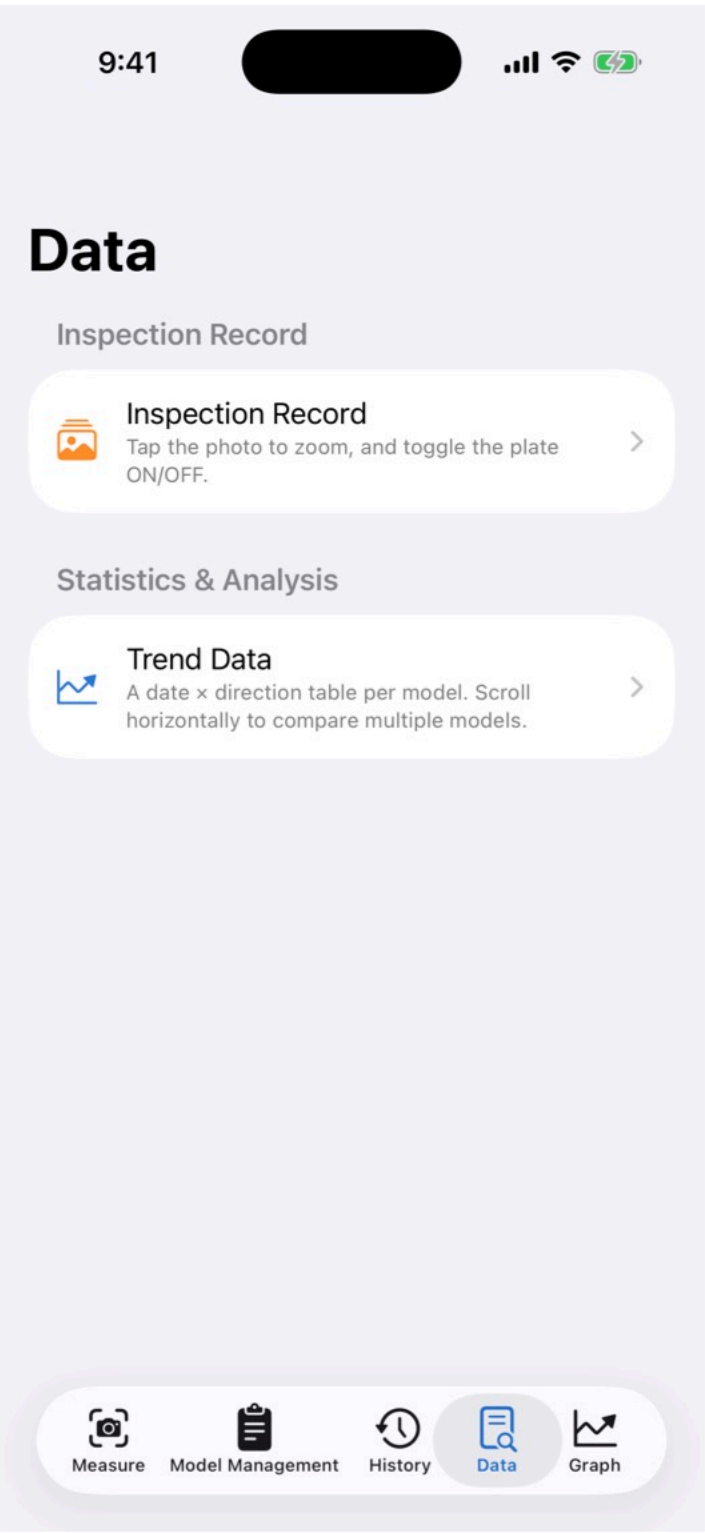
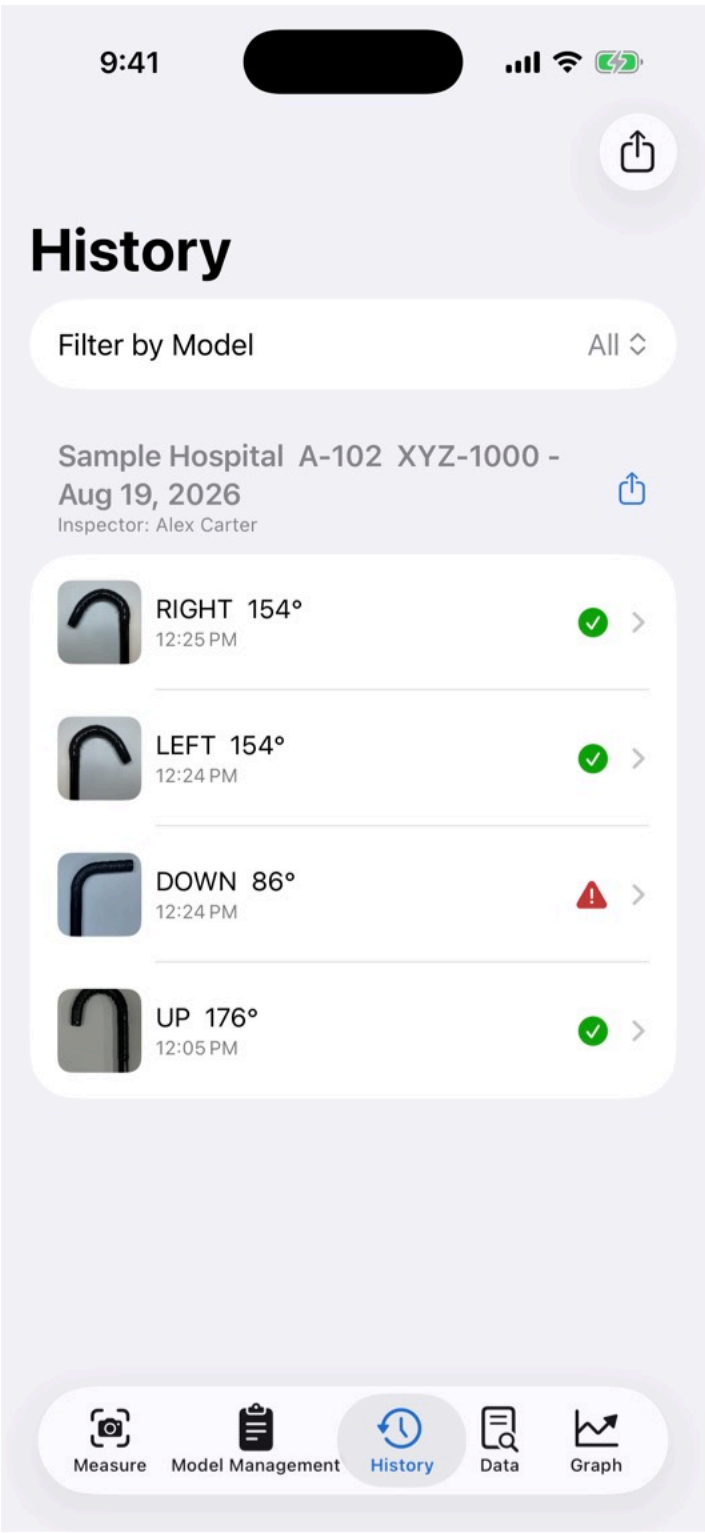
- Made for inspection work where both hands are busy: you can work by voice alone, without touching the screen.
- Voice control works only while the “Measure” tab is open.
- Change direction: “up” “down” “left” “right”
- Take a photo: “capture” “shoot” “photo”
- Save: “save”
- Shoot again: “back” “retry” “redo” “cancel”
- Zoom: “zoom in” “closer” / “zoom out” “farther”
- Focus: “focus”

07

Records and Reports

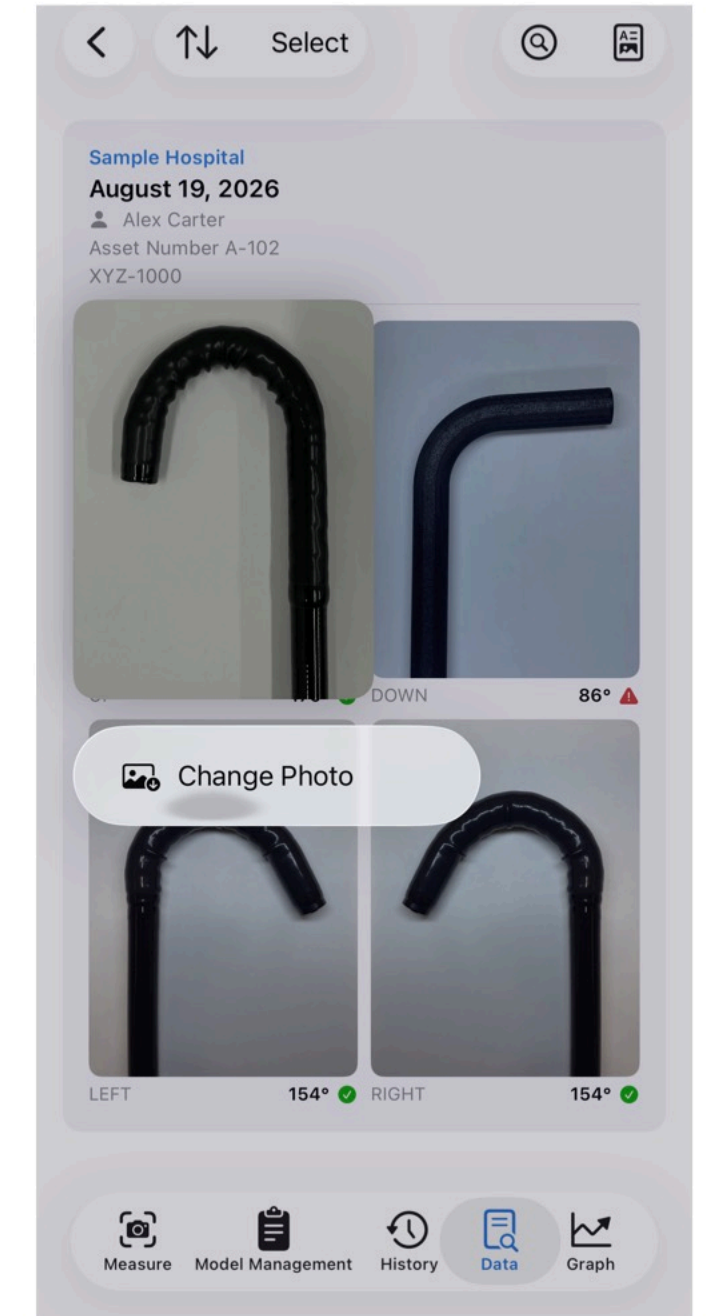
History / Data / Graph Tabs

- History tab: check, correct, and delete day-to-day records (for use on site)
- Data tab: browse and sort everything, and build reports (for monthly reporting and audits)
- Graph tab: see how the angle changes over time as a line graph. If the specified angle and tolerance are registered, the passing range is shown as a band.



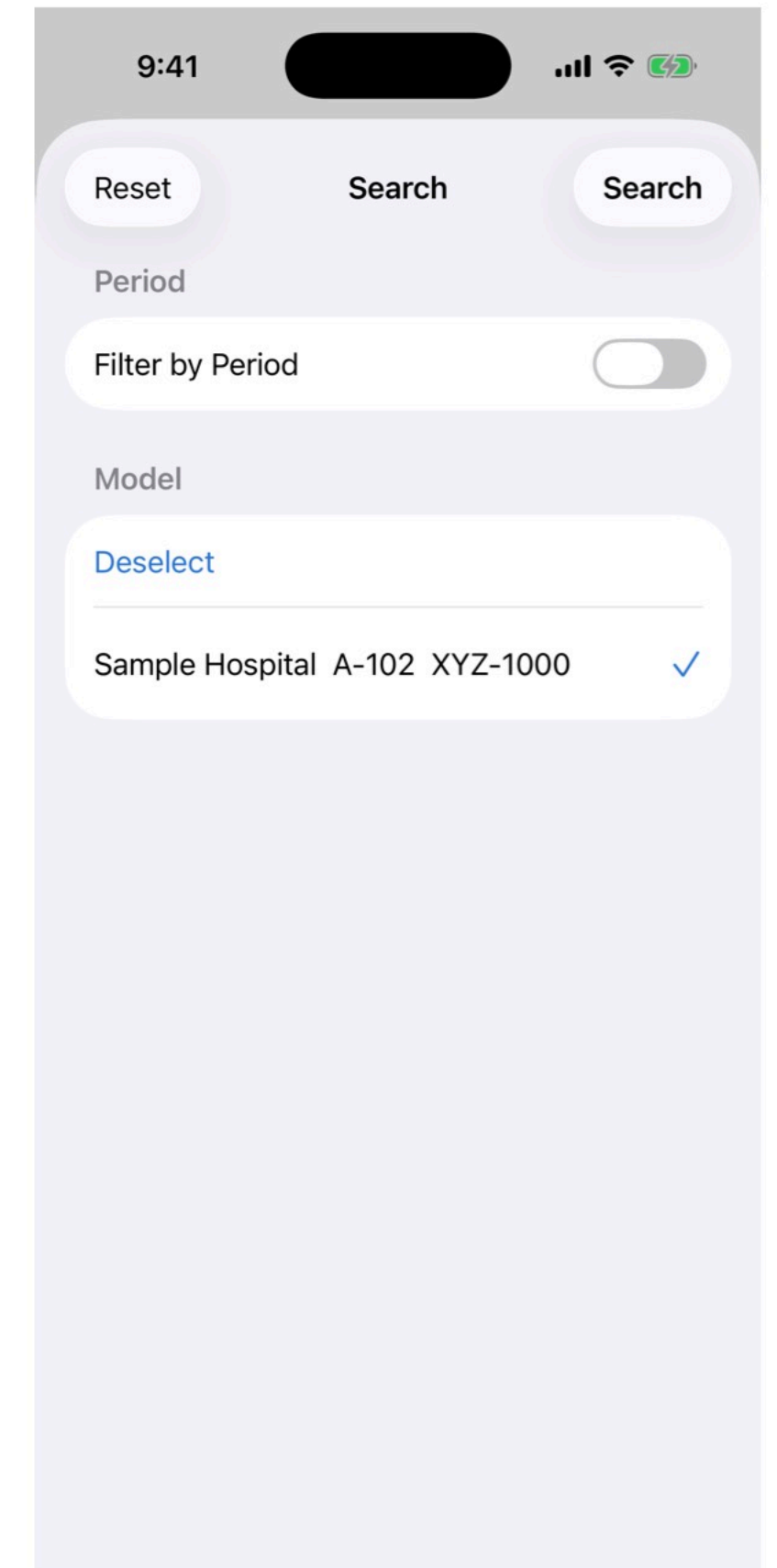
Viewing Records and Swapping Photos

- Data tab → “Inspection Record” lists the four directions' photos and angles for each model and date.
- Tap a photo to enlarge it (you can overlay the plate here too).
- Press and hold a photo → “Change Photo” to swap in another shot of the same direction.
- The candidates are listed with their time and measured angle, so you can pick the best one.



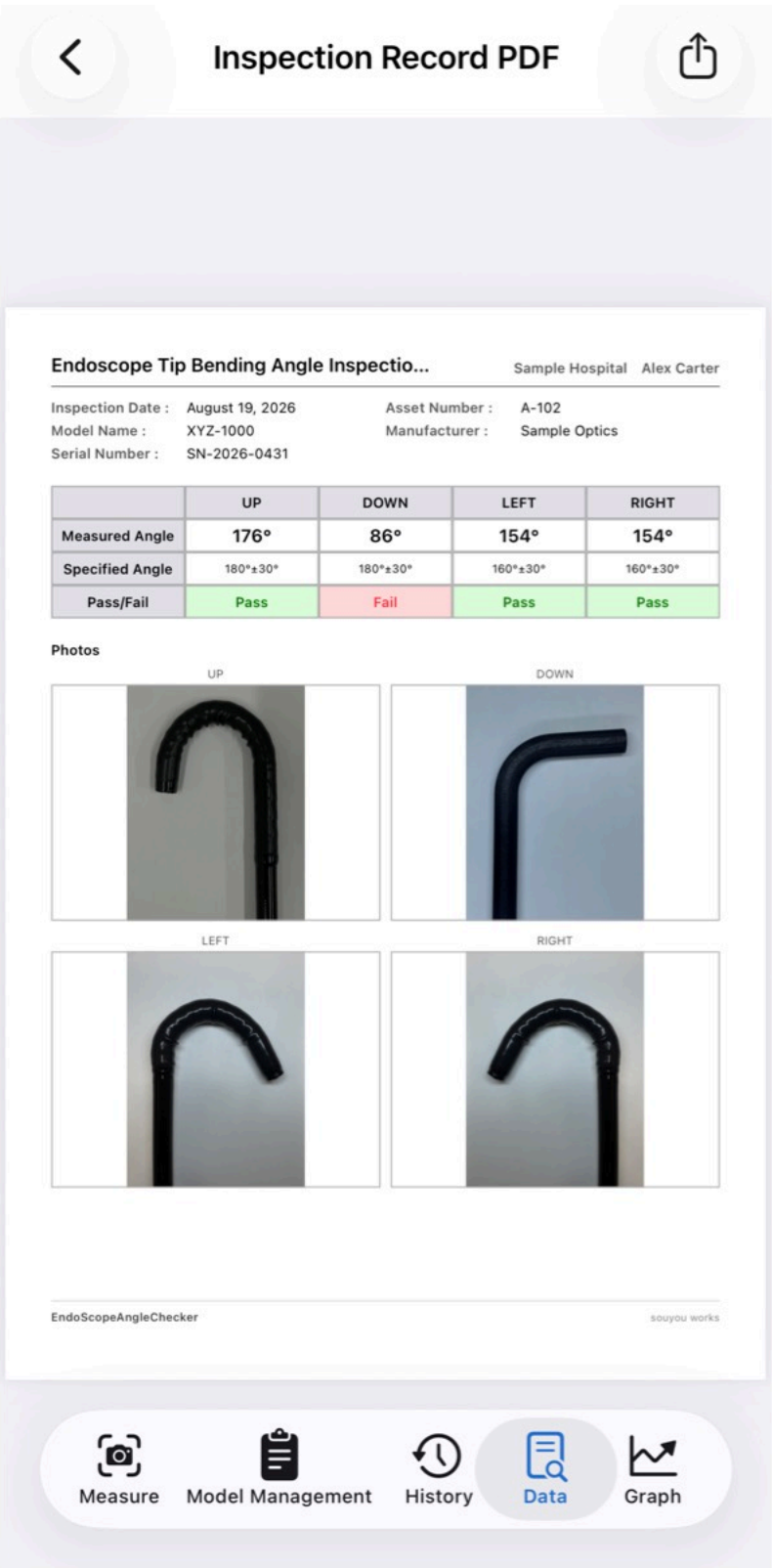
Searching and Filtering Records

- Use the search icon at the top of the inspection record list to filter.
- “Filter by Period”: set a range of inspection dates.
- “Model”: show only one model (by hospital and asset number).
- For monthly reports and audits, you can pull out just the records you need.



Exporting Reports as PDF and CSV

- Inspection Record PDF: one page per model and date, with a table of angle, specified angle, and pass/fail, plus the photos for all four directions
- Inspection Record CSV: one row per day, with the asset number, model name, and all four angles side by side
- Trend CSV: the four directions stacked vertically for each date (also ideal as a backup)
- Exporting opens the share sheet, so you can email it, AirDrop it, or save it to Files.
- As a regular backup, we recommend saving the Trend CSV or the Inspection Record PDF.

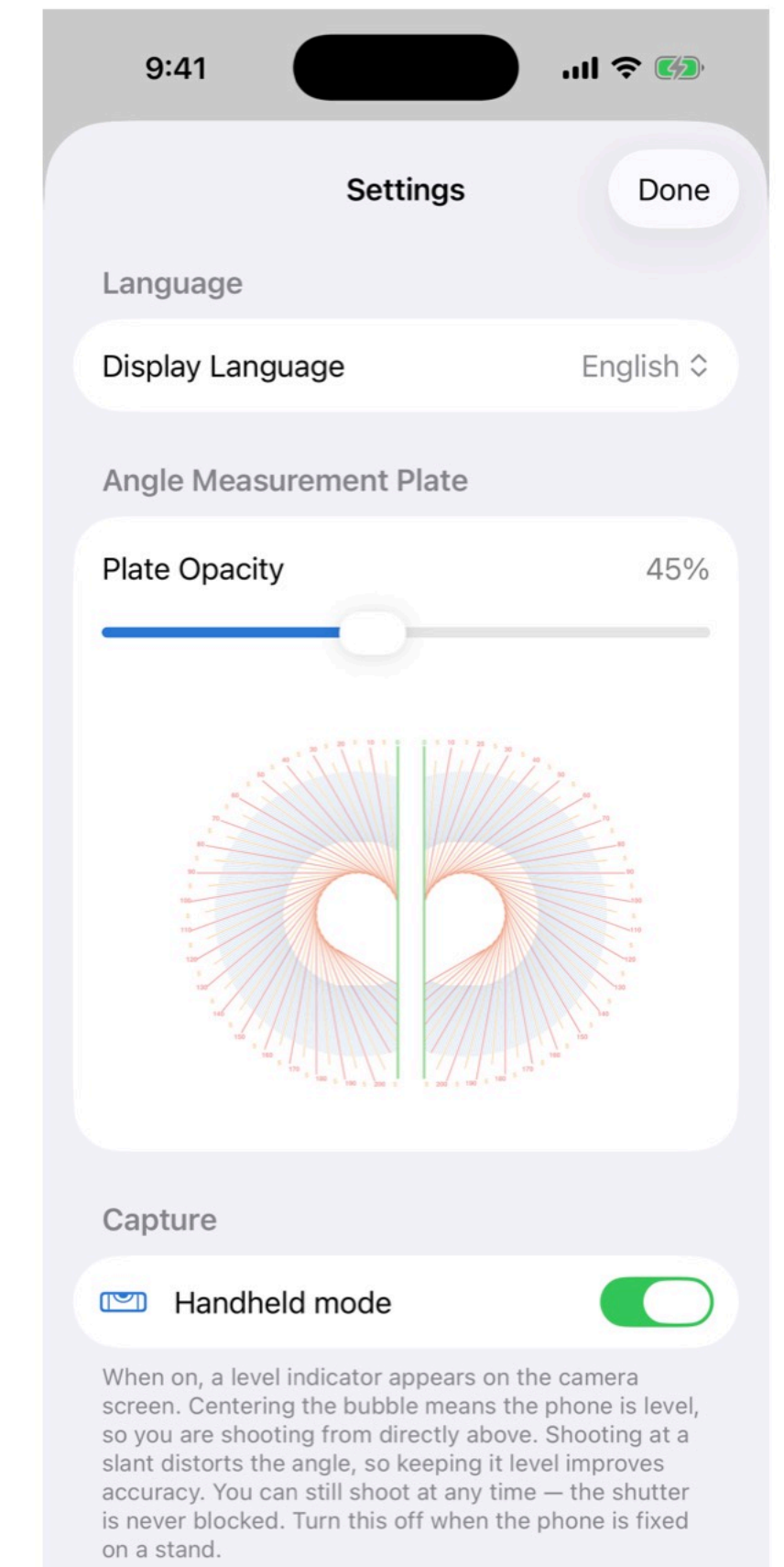


08

Settings

The Settings Screen

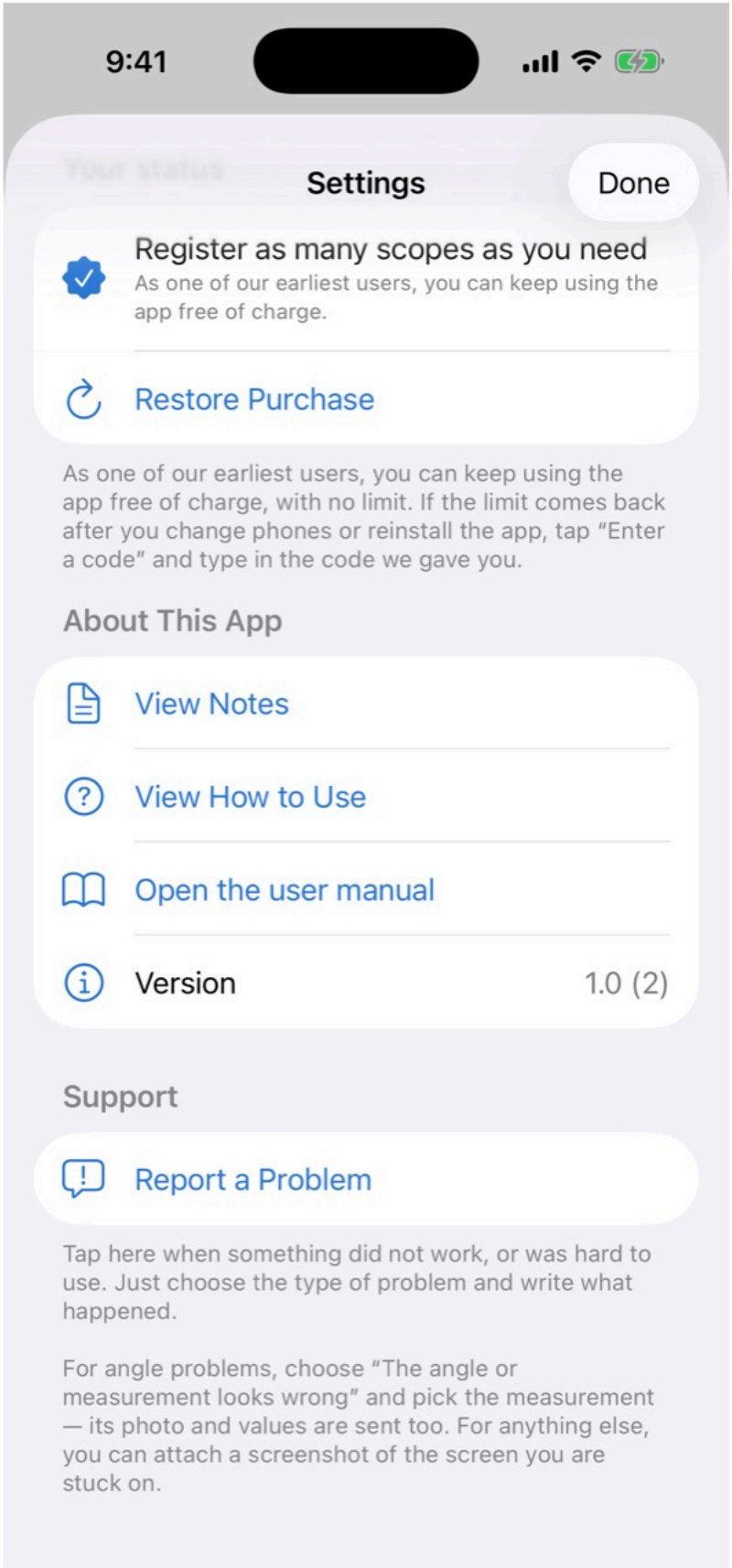
- Open it from the gear icon at the top right of the “Measure” tab.
- Display Language: 日本語 / English / 中文(简体)
- Plate Opacity: how strongly the angle plate is drawn (10%–90%)
- Handheld mode: turn it on to show a level indicator.
- Management: Models / Manage Hospitals / Manage Inspectors / Transfer Data



NOTE

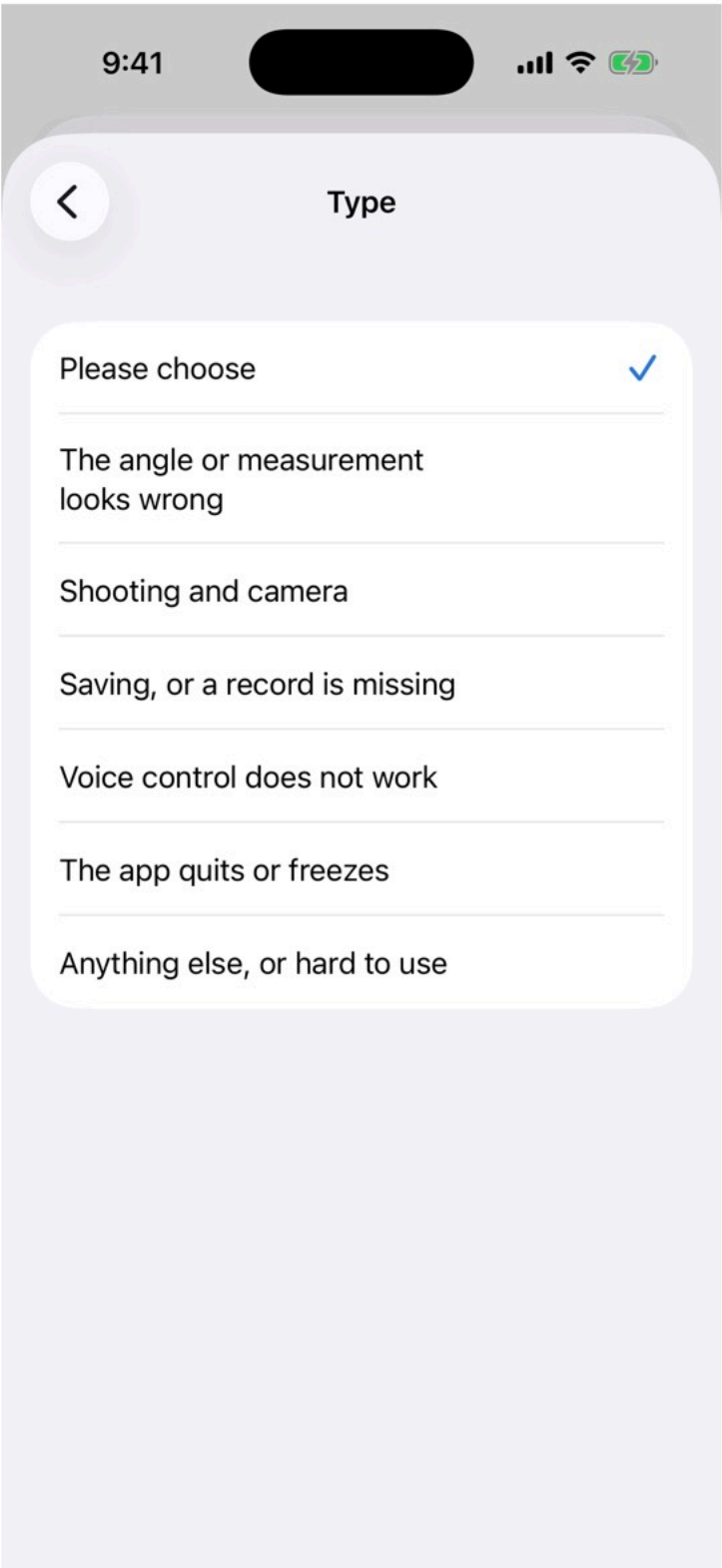
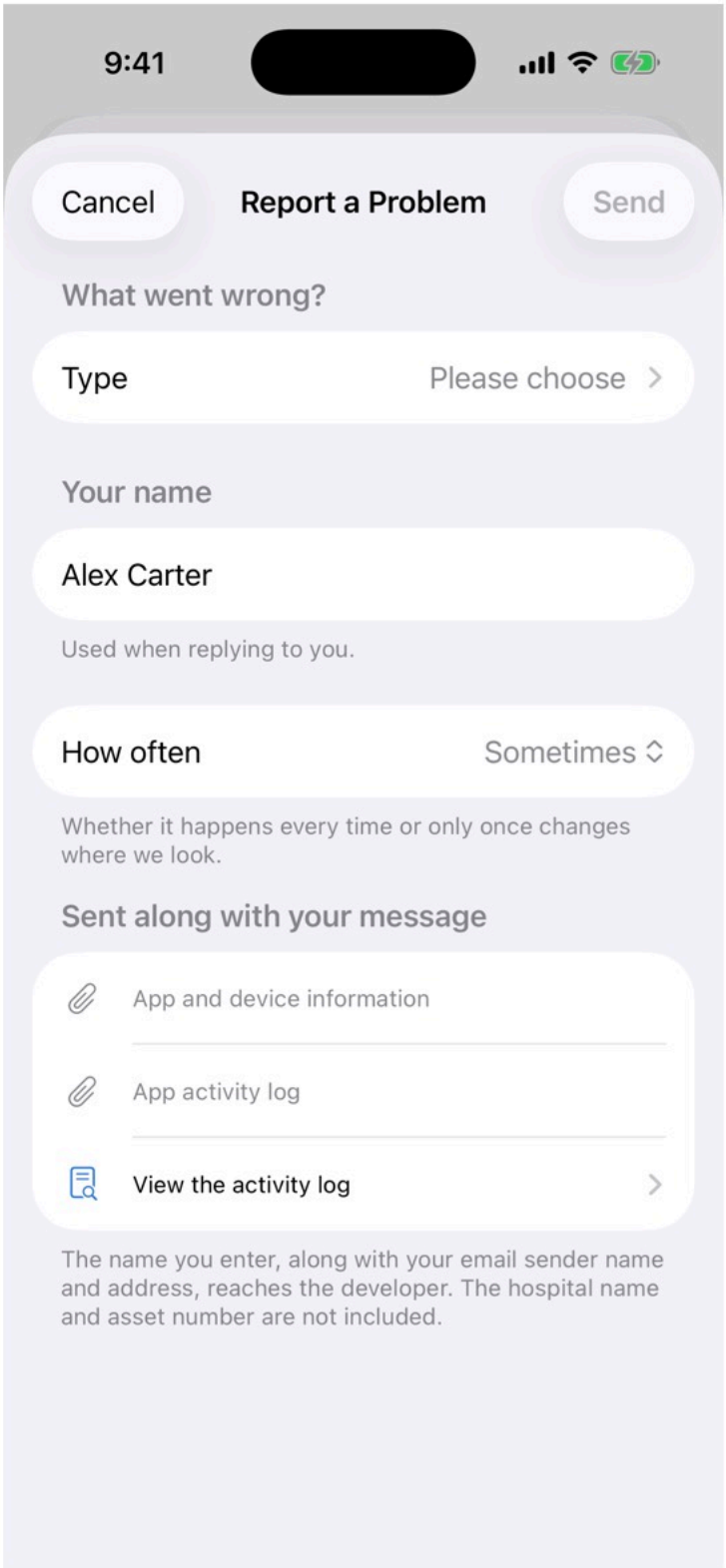
Other Settings

- About This App: open the notes, the how-to tutorial, and this manual at any time.
- Version: please quote this number when you report a problem.
- Report a Problem: choose a type, describe it, and add your name — the activity log is sent along with it.



Report a Problem

- Settings (gear) → “Report a Problem”. This is the only way in.
- 1. Choose a type (angle & measurement / camera / saving / voice / crashes / anything else).
- 2. Describe it. Examples of what to write appear below, for each type.
- 3. Enter your name. It is used when we reply.
- 4. Tap “Send”: a mail compose screen opens — just send it.



How to Attach Photos, and What Gets Sent

- Choosing “The angle or measurement looks wrong” reveals “Choose a measurement”. Pick the one in question and its photo and values (measured angle, specified angle, direction, model name) are sent too.
- If you would rather not send the photo, turn “Attach the measurement photo” off and report the values alone.
- For every other type you can attach a screenshot. To take one: press the side button and the volume up button together (on models with a Home button, press Home and the power button).
- What is sent: the type, what you wrote, your name, app and device information, and the app's activity log.
- What is not sent: hospital names, asset numbers, and your other measurement records.
- Because it goes by email, your email address and name do reach us (they are needed so we can reply).